VitalNav.AI

AI-Powered Emergency Triage & Navigation Platform

Care Begins Before Arrival.

A seamless AI platform that triages patients **in their own language**, routes them to the optimal emergency department, and delivers **actionable clinical intelligence in real time** to hospital staff — before the patient arrives.

DEVELOPED BY

Atamer Erkal

Healthcare AI Engineer

VitalNav.AI · MedTech · KI

~35%	~20 Min.	95%*	−28%*	DE · EN · TR	5-Level
Reduction in ED Overcrowding	Time Saved Before Triage	MTS Triage Accuracy *Target · Model Projection	Door-to-Treatment *Target · Model Projection	Languages (easily expandable)	MTS Clinical Triage Standard

THE PROBLEM — EVIDENCE-BASED

Emergency departments across Europe are **chronically overcrowded**. Germany recorded **13.0 million ambulatory emergency visits** in 2024 [1]. Health services research shows a significant share are medically non-urgent — a nationwide German analysis (*Health Policy*, 2023) confirms that **roughly half of ED visits are low-priority** [2]. Non-urgent cases consume resources, extend wait times, and endanger life-threatening patients.

A 10-year analysis (*BMC Psychiatry*, 2025) confirms the **continuous rise in non-acute emergency visits** [3]. **Language barriers** compound the crisis: migrants and tourists cannot accurately describe symptoms — **misclassification, delays and preventable harm** follow [4].

Hospitals have **zero advance visibility** into incoming patient load — beds, staff and specialists are deployed **reactively instead of proactively**.

[1] Destatis 2024 · [2] Health Policy 2023 · [3] BMC Psychiatry 2025 · [4] OECD/EU Reports on Language Barriers in Emergency Care

OUR SOLUTION

① **Multilingual AI Triage** — Patients describe symptoms via voice input or 12 symptom buttons, hands-free and in **their own language**. The AI applies the **MTS standard** and determines the urgency level.

② **Photo & Video Upload** — Visual pre-assessment improves diagnostic accuracy and prepares the team **before the patient arrives**.

③ **Smart Navigation** — Real-time capacity, specialist availability and traffic data determine the **optimal destination hospital**. The team receives symptom profile, triage level, AI recommendation & ETA.

④ **Expert-in-the-Loop** — Clinicians retain **full control**: every AI decision is reviewable, correctable and overridable. MDR & GDPR compliant.

THE VITALNAV.AI EFFECT — FROM EMERGENCY TO TREATMENT

95%* — **MTS Triage Accuracy**
**Target · Model Projection (pre-clinical validation)*

−28%* — **Door-to-Treatment Time**
**Target · Model Projection (pre-clinical validation)*

~20 Min. — **Earlier Clinical Insight**
Structured report before arrival

Traditional: Alarm → Transport (No data) → Hospital Triage → Bed Assignment → Treatment (Late) — Avg. +45 min delay

VitalNav.AI: Alarm → AI Triage at Home → Hospital prepares → Arrival = ready → Direct Treatment — −28% Zeit

CORE FEATURES

SPEAK & BE SAVED — AI Voice Triage (Core)
Complete symptom capture via voice or 12 buttons — no typing, hands-free, in any language. Adaptive clinical follow-up questions.

PRE-ARRIVAL TRIAGE — Real-Time MTS Classification (Core)
AI assigns MTS urgency level 1–5. Full clinical report delivered to the dashboard before the patient reaches the door.

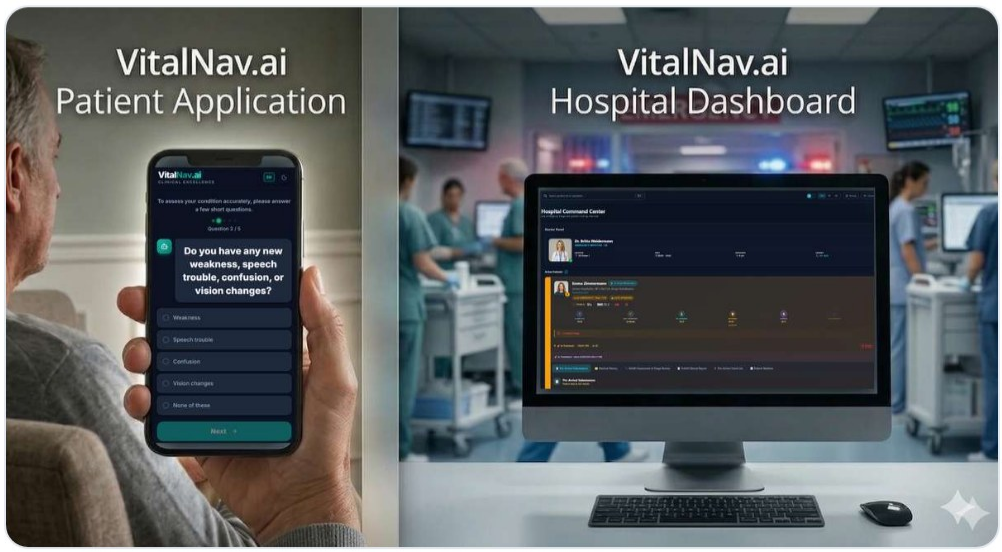
HUMAN IN CONTROL — Expert-in-the-Loop (Safety)
Clinician reviews, annotates and overrides every AI recommendation before clinical effect. MDR & GDPR compliant.

LIVE PHOTO & VIDEO — Visual Clinical Evidence (Core) (Unique)
Patient uploads photos/videos of injuries — during transport. Team sees the finding before the patient reaches the door.

OPTIMAL ER — Intelligent Multi-ER Routing (Core)
Compares multiple hospitals by real-time capacity & traffic. Patient is routed to the fastest, most appropriate emergency department.

FAMILY ALERT — One-Tap Emergency Notification (Critical)
Live GPS, status update & destination hospital are instantly shared with emergency contacts — no call, no speaking needed. Reduces chaos at the ED.

SILENT PANIC BUTTON — GPS Emergency Alert (Critical)
One tap instantly sends GPS to emergency services & destination hospital. No speaking required — life-saving when unable to act.



Patient App & Hospital Dashboard — one seamless platform

WHO BENEFITS & HOW

HOSPITAL BENEFITS

- ✓ **Structured data ~20 min before arrival** — pre-assign beds, alert specialists, proactively plan resources
- ✓ **~40% less admission administration** — nursing time flows into care, not paperwork
- ✓ **Real-time patient load insight** — overcrowding is prevented, not managed
- ✓ **AI bed & specialist recommendation** — right care, right place, from second one
- ✓ **No reception queue** — QR scan links data instantly on arrival
- ✓ **Full clinical oversight** — Expert-in-the-Loop keeps clinicians in control

PATIENT BENEFITS

- ✓ **Triage in own language in ~45 sec.** — voice or symptom button, no typing
- ✓ **Optimal emergency department** — not nearest, but fastest & most appropriate
- ✓ **No uncertainty en route** — live GPS, wait-time updates, doctor messages
- ✓ **Family automatically notified** — one tap, no call needed
- ✓ **Pre-arrival guidance** — DO/DON'T instructions in own language
- ✓ **Silent Panic Button** — GPS sent, help dispatched. No speaking.

ADDITIONAL FEATURES

QR Auto-Registration
Scan on arrival — data instantly linked to hospital record, zero manual entry

Bed & Doctor Assignment
AI recommends bed category & specialty based on triage level & ward capacity

Doctor→Patient Messaging
Voice messages are translated in real time into the patient's own language

112 / 116117 Direct Call
One-tap connection to emergency & on-call medical service — instant, no searching

Live ETA & Re-Triage
Arrival time updates dynamically; AI re-evaluates urgency if symptoms worsen during transport

Audit Trail & Logging
Every AI decision & clinician override is timestamped — forensically traceable, MDR compliant

Patient's Path
From symptoms at home to safe arrival

LIVE SYNC
Real-time data flow between two paths

Hospital's Path
From blind ED to a fully prepared team

Symptoms appear at home
Patient or family is suddenly facing an unclear emergency. **Panic, language barriers, no medical knowledge.** Calling a hotline feels intimidating.

ED operates blind
No incoming clinical data. **Beds, specialists and equipment are reactive**, not staged. The dark transit window has begun.

Opens VitalNav.AI — speaks own language
★ UNIQUE

Voice or text triage in **DE · EN · TR** (extendable). No medical jargon, no interpreter line, no hold music. App is ready 24/7.

Voice + Text · Multilingual

LIVE SYNC
Cross-lingual clinical bridge

Live data feed appears on dashboard
Patient's words arrive translated into a **structured clinical summary** in the clinician's language — instantly.

Auto-Translated · Structured Report

AI questions + photo/video upload
★ UNIQUE

Dynamic follow-ups adapt to symptom + history. Patient uploads **photos of burns, rashes, wounds** — visual evidence at the source.

MTS-Standard · Photo / Video

LIVE SYNC
Visual evidence before arrival

Triage team sees photos + MTS draft
Visual evidence + medical history + allergies arrive in real time. Up to **+40% triage accuracy** from photo evidence alone.

+40% Accuracy · Allergies + Meds

Receives clear, verified instructions
Patient is **informed in plain language**: what is happening, what to do next and where to go — *"Wait for the ambulance"* or *"Go to ER X — bed reserved for you."* No medical jargon. No guesswork.

VERIFY
HUMAN HAS FINAL WORD

Expert reviews · confirms · overrides
CRITICAL

Doctor or triage nurse sees the AI assessment and can confirm or override every decision. **No black box.** Every override is logged.

Expert-in-the-Loop · MDR-Compliant

Asks the team — in own language
★ UNIQUE

"Should I give him water?" · "He's vomiting again" — patient or family writes **in their native tongue**; the message is **auto-translated for the medical team** in real time.

Native Language · No Interpreter

AUTO-TRANSLATE
Two-way clinical chat

Replies — in own language
Doctor or triage nurse types in their **working language**; the patient receives the answer in their native one. **Clinical clarity, zero translation overhead.**

Real-Time · Clinical Chat

En route to the right ER
Family driving or ambulance dispatched. App keeps the patient updated, family informed, and **silent GPS panic** is one tap away.

Live GPS tracking · Silent panic button

GPS · 4S
Live ETA continuous update

Bed · specialist · equipment staged
★ UNIQUE

Smart ER selection picks the right hospital by **ER capacity, ER occupancy, real-time traffic and distance**. The receiving team is briefed before the doors open.

QR check-in at the door — no front desk
★ UNIQUE

Patient scans the QR shown in the app at the ER entrance. **No paperwork, no queue, no language confusion.** Identity, insurance and clinical record are confirmed in seconds.

QR Fast-Lane · No Paperwork

AUTO CHECK-IN
Zero registration delay

Routed directly to the assigned room
App guides the patient straight to the **pre-assigned room and doctor**. The triage desk is bypassed entirely — minutes saved are minutes earlier of treatment.

Direct to Doctor · Indoor Wayfinding

THE TWO PATHS MEET · TREATMENT BEGINS

Patient walks in already known — the team is already prepared.

DOOR-TO-TREATMENT
−28%*
*model projection

ONLY VITALNAV.AI

- Multilingual voice triage
- Two-way auto-translated chat
- Photo / video clinical evidence
- MTS-standard triage 1–5
- Expert-in-the-Loop
- Smart ER selection (capacity · occupancy · traffic · distance)
- 4-second GPS sync
- Live patient tracking
- Silent GPS panic
- QR fast-lane · direct to doctor
- Emergency contact notification
- Pre-hospital data package to ER

Complementing 112 / 116117 — the Pre-Arrival Clinical Data Layer for Hospitals

112 & 116117 — ESSENTIAL PUBLIC SERVICE
Life-saving voice dispatch and on-call routing. Irreplaceable infrastructure for true emergencies.

VITALNAV.AI — THE VISUAL & DATA BRIDGE
*Unlike voice-only services, we don't just **call** the hospital — we **prepare** it. MTS-graded symptoms, photos, history, allergies and a multilingual clinical report delivered before arrival.*

The Multilingual Clinical Bridge

Patient speaks Turkish — clinician reads an English/German clinical summary. **VitalNav.AI is the only platform** that closes the dangerous language gap voice hotlines structurally cannot solve.

Visual Evidence Decides Triage

Burns, rashes, paediatric trauma — photo/video at the source can boost triage accuracy by up to **40%**. **Only VitalNav.AI** sends visual evidence to the triage team **before arrival**.

The Clinician Knows the Patient Before the Door Opens

~20 min pre-arrival report: MTS level, complaint, photos, voice transcript, **medical history, allergies and current medications**. Bed, specialist and equipment staged proactively.

VS. 112 / 116117

What hotlines structurally cannot deliver

Capability	VitalNav.AI ★	112	116117
Structured pre-arrival clinical data (~20 min lead)	✓	✗	✗
Photo / video evidence of the finding	✓	✗	✗
Cross-lingual clinical summary & communication (patient language ↔ clinician language)	✓	✗ interpreter line	✗ DE only
Medical history, allergies & medications captured at triage	✓	✗	✗
AI questions adapted to symptom + medical history	✓	~ AMPDS protocol	~ SmED protocol
Algorithmic MTS triage (level 1–5, deterministic)	✓	~ dispatcher-led	~ SmED-led
Multi-ER routing by capacity + ETA + traffic	✓	✗ nearest hospital	✗
Live hospital dashboard with continuous updates	✓	✗	✗
GPS update every ~4s during transport + dynamic ETA	✓	~ dispatch only	✗
Silent GPS emergency alert (no speaking required)	✓	✗ voice required	✗
Forensic audit trail for every AI decision & override	✓	~ call recording	~ call recording

VS. OTHER PLATFORMS

Market comparison · Apr. 2026

Capability	VitalNav.AI ★	Inferm-edica	Ada Health	Docto-lib
AI symptom triage (pre-clinical)	✓	✓ API	✓	✗
MTS-standard triage (level 1–5)	✓	✗ own scoring	✗ own scoring	✗
Photo / video symptom evidence	✓	✗	✗	✗
Multi-ER real-time routing (capacity + ETA)	✓	✗	✗	✗
Pre-arrival hospital dashboard (ED-grade)	✓	✗	✗	~ clinic portal
Cross-lingual clinical summary & communication	✓	✗	✗	✗
AI bed & specialist assignment	✓	✗	✗	✗
Silent GPS panic button (emergency-grade)	✓	✗	✗	✗
Multilingual symptom input	✓	✓ 20+ langs	✓ 12+ langs	✓ EU langs
Voice-driven symptom capture	✓	~ via partners	~ limited	✗
CE-MDR / market approval	~ in prep.	✓	✓	✓
HIS/EHR integration depth	~ FHIR pilot	✓	~	✓

✓ Fully available · ~ Partial · ✗ Not available · ★ VitalNav.AI · Apr. 2026

TECH STACK GPT-5.4 Mini · Azure AI Search (RAG) · Deterministic Rules · MTS Algorithm · FastAPI / Python · FHIR-ready API · WebSocket Live Feed · SQLite (encrypted)

Expert-in-the-Loop

MDR · GDPR · MTS Standard

VitalNav.AI
Care Begins Before Arrival

The **first end-to-end platform** uniting pre-hospital voice triage, visual clinical evidence, multilingual AI reasoning, full medical-history capture, and real-time multi-ER routing in a single regulation-ready system — the structured-data layer that **illuminates the dark transit window** between 112 dispatch and hospital arrival.